

Unit - 3

Numerical Differentiation And Integration:

① Derivative using Newton's Forward Interpolation formulae.

$$P = \frac{x - x_0}{h}$$

$$y = y_0 + P \Delta y_0 + \frac{P(P-1)}{2!} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3!} \Delta^3 y_0 + \frac{P(P-1)(P-2)(P-3)}{4!} \Delta^4 y_0 + \dots$$

$$\frac{dy}{dx} = \frac{dy}{dP} \cdot \frac{dP}{dx} \quad \text{--- (I)}$$

$$\boxed{\frac{dP}{dx} = \frac{1}{h}} \quad \text{--- (II)}$$

$$\frac{dy}{dP} = 0 + \Delta y_0 + \frac{2P-1}{2} \Delta^2 y_0 + \frac{3P^2-6P+2}{6} \Delta^3 y_0 + \frac{4P^3-18P^2+22P-6}{24} \Delta^4 y_0 + \dots \quad \text{--- (III)}$$

Putting (II) & (III) in (I)

$$\left\{ \frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{2P-1}{2} \Delta^2 y_0 + \frac{3P^2-6P+2}{6} \Delta^3 y_0 + \frac{4P^3-18P^2+22P-6}{24} \Delta^4 y_0 + \dots \right] \right\} \quad \text{--- first derivative.}$$

Now, at $x = x_0$, $P = \frac{x - x_0}{h} = \frac{x_0 - x_0}{h} = 0$

$$\boxed{P=0}$$

$$\left\{ \frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right] \right\}$$

• 2nd derivative. ... By derivating four derivation.

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{dy}{dp} \cdot \frac{dp}{dx} \right) = \frac{d^2y}{dp^2} \cdot \frac{dp}{dx} \cdot \frac{dp}{dx}$$

$$\frac{d^2y}{dp^2} = 0 + \Delta^2 y_0 + \frac{6P-6}{6} \Delta^3 y_0 + \frac{12P^2-36P+22}{24} \Delta^4 y_0 + \dots$$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2} \left[\Delta^2 y_0 + \frac{6P-6}{6} \Delta^3 y_0 + \frac{12P^2-36P+22}{24} \Delta^4 y_0 + \dots \right]$$

→ Now, at $x=x_0$, $P=0$

$$\left(\frac{d^2y}{dx^2} \right)_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \frac{137}{120} \Delta^6 y_0 + \dots \right]$$

(ii) Derivating using Newton's Backward Formulae:

$$y = y_0 + P \nabla y_0$$

$$y = y_0 + P \nabla y_0 + \frac{P(P+1)}{2!} \nabla^2 y_0 + \frac{P(P+1)(P+2)}{3!} \nabla^3 y_0 + \dots$$

$$\frac{d^2y}{dx^2} = \frac{1}{h} \left[\nabla^2 y_0 + \frac{2P+1}{2} \nabla^3 y_0 + \frac{3P^2+6P+2}{6} \nabla^4 y_0 + \frac{4P^3+18P^2+22P+6}{24} \nabla^5 y_0 + \dots \right]$$

→ at $x=x_0$, $P=0$.

$$\frac{d^2y}{dx^2} = \frac{1}{h} \left[\nabla^2 y_0 + \frac{1}{2} \nabla^3 y_0 + \frac{1}{3} \nabla^4 y_0 + \frac{1}{4} \nabla^5 y_0 + \dots \right]$$

(ii) Derivating using Newton's Backward Interpolation

$$P = \frac{x - x_n}{h}$$

$$y = y_n + P \nabla y_n + \frac{P(P+1)}{2!} \nabla^2 y_n + \frac{P(P+1)(P+2)}{3!} \nabla^3 y_n + \frac{P(P+1)(P+2)(P+3)}{4!} \nabla^4 y_n$$

+

$$\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \frac{2P+1}{2} \nabla^2 y_n + \frac{3P^2+6P+2}{6} \nabla^3 y_n + \frac{4P^3+18P^2+22P+6}{24} \nabla^4 y_n + \dots \right]$$

→ at $x = x_n$

$$\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n + \dots \right]$$

→ Second derivative :

$$\frac{d^2y}{dx^2} = \frac{1}{h^2} \left[\nabla^2 y_n + \frac{6P+6}{6} \nabla^3 y_n + \frac{12P^2+36P+22}{24} \nabla^4 y_n + \dots \right]$$

→ at $x = x_n$ $P = 0$

$$\frac{d^2y}{dx^2} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_n + \frac{5}{6} \nabla^5 y_n + \dots \right]$$

⇒ Numerical Integration:

$$I = \int_{x_0}^{x_n} f(x) \cdot dx$$

$$h = \frac{x_n - x_0}{n}$$

where n = no. of intervals

⇒ Using Newton's forward interpolation formulae.
Newton's Cotes quadrature formulae.

$$I = h \left[n y_0 + \frac{n^2}{2} \Delta y_0 \right] \rightarrow \text{when } n=1$$

$$I = h \left[n y_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2!} + \left(\frac{n^4}{2} - n^3 + n^2 \right) \frac{\Delta^3 y_0}{3!} + \dots \right]$$

Rules:

① Trapezoidal Rule: $n=1$, n should be a multiple of 1.

$$\Delta y_0 = y_1 - y_0$$

$$\Delta y_1 = y_2 - y_1$$

$$I = \int_{x_0}^{x_1} f(x) \cdot dx = h \left[y_0 + \frac{1}{2} (y_1 - y_0) \right]$$

$$= \frac{h}{2} [2y_0 + y_1 - y_0] = \frac{h}{2} [y_1 + y_0]$$

When no. of intervals = 1

i.e.

x	y
x_0	y_0
$x_0 + h$	y_1

$$\rightarrow \Delta y_0 = y_1 - y_0$$

In this we take n (i.e. no. of intervals = 6) as a standard.

Similarly,

$$I = \int_{x_1}^{x_2} f(x) \cdot dx = h \left[y_1 + \frac{1}{2}(y_2 - y_1) \right]$$

$$= \frac{h}{2} [y_2 + y_1]$$

...

$$I = \int_{x_{n-1}}^{x_n} f(x) dx = \frac{h}{2} [y_{n-1} + y_n]$$

~~is added~~
2.

$$I = \int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [y_0 + y_1 + y_2 + y_2 + y_2 + \dots + y_{n-1} + y_{n-1} + y_n]$$

$$= \frac{h}{2} [y_0 + 2y_1 + 2y_2 + 2y_3 + \dots + 2y_{n-1} + y_n]$$

$$I = \int_{x_0}^{x_n} f(x) dx = \frac{h}{2} [(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1})]$$

Table	
x	y
x_0	y_0
x_0+h	y_1
x_0+2h	y_2
x_0+3h	y_3
$\vdots$	$\vdots$
x_0+nh	y_n

② Simpson's $\frac{1}{3}$ rule. ($n=2$).

$$I = \frac{h}{3} [(y_0 + y_n) + 2(y_2 + y_4 + \dots + y_{n-2}) + 4(y_1 + y_3 + y_5 + \dots + y_{n-1})]$$

to learn easily.

$$I = \frac{h}{3} [(y_0 + y_n) + 2(\text{no. of even terms}) + 4(\text{no. of odd terms})]$$

x	y
x_0	y_0
x_0+h	y_1
x_0+2h	y_2
x_0+3h	y_3
x_0+4h	y_4
$\vdots$	$\vdots$
x_0+nh	y_n

no. of intervals = 6 as an standard.

x

③ Simpson's $\frac{3}{8}$ rule: $n=3$

x

x_0
 x_0+h
 x_0+2h
 x_0+3h
 x_0+4h
 x_0+5h
 x_0+6h
 $\vdots$
 x_0+nh

$$I = \frac{3h}{8} \left[(y_0 + y_n) + 2(y_3 + y_6 + y_9 + \dots + y_{n-3}) + 3(y_1 + y_2 + y_4 + \dots + y_{n-2} + y_{n-1}) \right]$$

to learn:

$$I = \frac{3h}{8} \left[(y_0 + y_n) + 2(\text{multiples of } 3) + 3(\text{Remaining terms}) \right]$$

④ Boole's Rule: $n=4$

x

x_0
 x_0+h
 x_0+2h
 x_0+3h
 x_0+4h
 x_0+5h
 x_0+6h
 x_0+7h
 x_0+8h
 x_0+9h
 $\vdots$
 x_0+nh

~~where~~ In case of 1 interval ($y_0 - y_4$)

$$I = \frac{2h}{45} \left[7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4 \right]$$

$$I = \frac{2h}{45} \left[7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4 + 7y_4 + 32y_5 + 12y_6 + 32y_7 + 7y_8 + 7y_8 + \dots + 32y_{n-3} + 12y_{n-2} + 32y_{n-1} + 7y_n \right]$$

$$I = \frac{2h}{45} \left[7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 + 32y_5 + 12y_6 + 32y_7 + 14y_8 + \dots + 32y_{n-3} + 12y_{n-2} + 32y_{n-1} + 7y_n \right]$$

• +855 29

⑤ Weddle's Rule (n = 6)

$$I = \frac{3h}{10} \left(y_0 + 5y_1 + 8y_2 + 6y_3 + y_4 + 5y_5 + y_6 \right)$$

0-6
6-12

$$I = \frac{3h}{10} \left[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12} \right]$$

$$I = \frac{3 \times 1}{10 \times 2} \left[1 + 5(.972973) + (.9) + 6(.82) + (.692308) + 5(.590164) + .5 \right]$$

$$= .785399$$

Solution of Differential Equation of Order 1:
The rate of change
of one variable on
other.

$$\frac{dy}{dx} = f(x, y) \quad \text{--- (1)}$$

$$y = f(x)$$

$$y(x) = ?$$

$$y(x_0) = y_0$$

Initial / Boundary Condition.

- ① Euler's Method Taylor Series Method
- ② Picard's Method.
- ③ Euler method.
- ④ Modified Euler method.
- ⑤ R-K Method

General Method

$$\frac{dy}{dx} + Py = Q$$

Integration factor:

$$I.F = e^{\int P dx}$$

Find Solution:

$$y \cdot I.F = \int Q \cdot I.F dx + C$$

① Taylor's Series Method:

$$\frac{dy}{dx} = f(x, y) = y'$$

$$= y'(x_0)$$

$$\frac{d^2y}{dx^2}$$

$$= y''$$

$$= y''(x_0)$$

$$\frac{d^ny}{dx^n}$$

$$= y^{(n)}$$

$$y^{(n)}(x_0)$$

Taylor Series

$$y = y_0 + (x - x_0) y'(x_0) + \frac{(x - x_0)^2}{2!} y''(x_0) + \frac{(x - x_0)^3}{3!} y'''(x_0) + \dots$$

$$\int f_1 \cdot f_2 dx = f_1 \int f_2 dx - \int \left[\frac{d}{dx} f_1 \int f_2 dx \right] dx + C$$

Ques: Using Taylor's series method obtain the value of y at $x = 0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$. Compare the numerical solution obtained with the exact solution.

$y_0 = 0$, $x = 0$ [till 4 to 5 steps]

$$\frac{dy}{dx} = 2y + 3e^x$$

$$y'(0) = 2 \times 0 + 3e^0 = 3$$

$$y'' = 2y' + 3e^x$$

$$y''(0) = 2 \times 3 + 3e^0 = 9$$

$$y''' = 2y'' + 3e^x$$

$$y'''(0) = 2 \times 9 + 3e^0 = 21$$

$$y^{(4)} = 2y''' + 3e^x$$

$$y^{(4)}(0) = 2 \times 21 + 3e^0 = 45$$

$$y^{(5)} = 2y^{(4)} + 3e^x$$

$$y^{(5)}(0) = 2 \times 45 + 3e^0 = 93$$

$$y(x) = 0 + \frac{(x-0)^1}{1!} \times 3 + \frac{(x-0)^2}{2!} \times 9 + \frac{(x-0)^3}{3!} \times 21 + \frac{(x-0)^4}{4!} \times 45 + \frac{(x-0)^5}{5!} \times 93$$

$$= 0 + (.2) \times 3 + \frac{(.2)^2}{2} \times 9 + \frac{(.2)^3}{6} \times 21 + \frac{(.2)^4}{24} \times 45 + \frac{(.2)^5}{120} \times 93$$

$$= 0 + .6 + .18 + 0.028 + 0.003 + 0.000248$$

$$= 0.721248 = 0.811248 \text{ Ans}$$

checking comparing with General.

$$\frac{dy}{dx} = 2y + 3e^x$$

$$\frac{dy}{dx} - 2y = 3e^x$$

$$P = -2$$

$$Q = 3e^x$$

$$I f = e^{\int P dx} = e^{-2x}$$

final:

$$y \cdot I f = \int Q \cdot I f dx + C$$

$$y \cdot e^{-2x} = \int 3e^x \cdot e^{-2x} dx + C$$

$$y e^{-2x} = 3 \int e^{-x} dx + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$x^0 = \frac{x^{0+1}}{0+1} = x$$

$$\int e^{ax} = \frac{e^{ax}}{a}$$

$$y e^{-2x} =$$

$$y = \frac{1}{e}$$

$$y = -$$

$$y = 0$$

$$y =$$

$$(1) \int_4^5$$

(iii) The Find 1961

ye
Popula
ye

$$y e^{-2x} = -3e^{-x} + C$$

$$y = \frac{1}{e^{-2x}} [-3e^{-x} + C] = -e^{2x} \cdot 3e^{-x} + C e^{2x}$$

$$y = -3e^x + C e^{2x}$$

$$y=0, x=0$$

$$0 = -3e^0 + C e^0$$

$$0 = -3 + C$$

$$C = 3$$

$$y = -3e^x + 3e^{2x}$$

$$y = -3e^x + 3e^{2x} = -3e^0 + 3e^{(0.4)}$$

$$y(0.2) = -3e^{(0.2)} + 3e^{(0.4)}$$

$$y(0.2) = 0.811265$$

Assignment - NO.3

① $\int_4^{5.2} \log_e x \cdot dx$ $\frac{u}{5 \text{ method}}$ $n=12$

② $\int_0^1 e^{-x^2} dx$

all method $n=12$

③ The population of a certain town is given below.
Find the rate of growth of population in 1931, 1941, 1961, & 1971. forward

Year(x):	1931	1941	1951	1961	1971
Population in year(y):	40.62	60.80	79.95	103.56	132.65

② Picard's Method

initial condition.

$$y(x_0) = y_0$$

$$\frac{dy}{dx} = f(x, y)$$

$$dy = f(x, y) dx$$

at least 5 times.

$$\int_{y_0}^y dy = \int_{x_0}^x f(x, y) dx$$

$$[y]_{y_0}^y = \int_{x_0}^x f(x, y_0) dx$$

$$y - y_0 = \int_{x_0}^x f(x, y_0) dx$$

$$y = y_0 + \int_{x_0}^x f(x, y_0) dx$$

let say it y_1

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$$

Similarly.

$$y_2 = y_0 + \int_{x_0}^x f(x, y_1) dx$$

⋮

$$\underline{y}_n = y_0 + \int_{x_0}^x f(x, \underline{y}_{n-1}) dx$$

$$\text{Ques } \frac{dy}{dx} = 2y + 3e^x$$

$$y(0) = 0, y(0.2)$$

$$y_1 = y_0 + \int_0^x (2y_0 + 3e^x) dx$$

$$y_1 = 0 + \int_0^x 3e^x dx = [3e^x]_0^x = 3e^x - 3e^0$$

$$y_1 = 3e^x - 3$$

$$y_1 = 3e^x - 3$$

$$y_2 = 0 + \int_0^x (2(3e^x - 3) + 3e^x) dx$$

$$y_2 = \int_0^x (6e^x - 6 + 3e^x) dx = \int_0^x (9e^x - 6) dx$$

$$= [9e^x - 6x]_0^x = (9e^x - 6x) - (9e^0 - 6 \cdot 0)$$

$$y_2 = 9e^x - 6x - 9$$

$$y_3 = 0 + \int_0^x (2(9e^x - 6x - 9) + 3e^x) dx$$

$$= 0 + \int_0^x (18e^x - 12x - 18 + 3e^x) dx$$

$$= 0 + \int_0^x (21e^x - 12x - 18) dx$$

$$= [21e^x - \frac{12x^2}{2} - 18x]_0^x = 21e^x - 6x^2 - 18x - 21$$

$$y_3 = 21e^x - 6x^2 - 18x - 21$$

$$y_4 = 0 + \int_0^x (21e^x - 6x^2 - 18x - 21) dx + 3e^x$$

$$y_4 = \left[45e^x - \frac{4}{3}x^3 - \frac{18}{2}x^2 - 45x \right]_0^x$$

$$y_4 = 45e^x - 4x^3 - 18x^2 - 45x - 45$$

$$y_5 = 0 + \int_0^x (90e^x - 8x^3 - 36x^2 - 90x - 90 + 3e^x) dx$$

$$= \left[93e^x - \frac{2}{4}x^4 - \frac{12}{3}x^3 - \frac{36}{2}x^2 - 90x - 90 \right]_0^x$$

$$y_5 = 93e^x - 2x^4 - 12x^3 - 42x^2 - 90x - 93$$

$$y = y_5 \text{ (let)}$$

$$= 93e^x - 2x^4 - 12x^3 - 42x^2 - 90x - 93$$

$$y(2) = 93e^2$$

$$= 113.590456 - 0.0032 - 0.096 - 1.68 - 18 - 93$$

$$y(2) = \underline{\underline{0.811256}}$$

→ Euler's Method

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0, \quad y(x) = ?$$

limit $x_0 \rightarrow x$ will be divided into unequal parts.

drawback:

it generate more error.

$$h = \frac{x - x_0}{n}$$

$$y_1 = y_0 + h f(x_0, y_0)$$

$$y_2 = y_1 + h f(x_1, y_1)$$

⋮

$$[y_n = y_{n-1} + h f(x_{n-1}, y_{n-1})]$$

(let $n=5$)

$$\frac{dy}{dx} = 2y + 3e^x$$

$$, y(0) = 0, y(0.2) = ?$$

$$h = \frac{0.2 - 0}{5} = 0.04$$

$$y_1 = 0 + 0.04(2(0) + 3e^0) = 0.12$$

$$y_2 = 0.12 + 0.04(2(0.12) + 3e^{0.04}) = 0.254497$$

$$y_3 = 0.254497 + 0.04(2(0.254497) + 3e^{0.08}) = 0.404851$$

$$y_4 = 0.404851 + 0.04(2(0.404851) + 3e^{0.12}) = 0.572539$$

x

0.00

0.04

0.08

0.12

0.16

0.20

— 0.20 —

$$y_5 = 0.572539 + 0.04(2 \times (0.572539) + 3e^{.16})$$

$$= .759163 \underline{\underline{Ans.}}$$

using Euler's Method.

Ques $\frac{dy}{dx} = \log_{10}(x+y)$

$y=1$

$x=0$

$h=0.1$

Sol:

$n=5$

$h = \frac{x-x_0}{n} \Rightarrow .1 = \frac{0-x_0}{5} \Rightarrow x_0 = 0$

x

$0 \ x_0$

$.1 \ x_0+h$

$.2 \ x_0+2h$

$.3 \ x_0+3h$

$.4$

$.5$

$y_1 = y_0 + hf(x_0, y_0)$

$y_1 = 1 + .1[\log_{10}(0+1)]$

$y_1 = 1$

$y_2 = 1 + .1(\log_{10}(.1+1))$

$y_2 = 1.004139$

$y_3 = 1.004139 + .1(\log_{10}(.2+1.004139))$

$y_3 = 1.012207$

$y_4 = 1.012207 + .1(\log_{10}(.3+1.012207))$

$y_4 = 1.024007$

$y_5 = 1.024007 + .1(\log_{10}(.4+1.024007))$

$y_5 = 1.039358$

⇒ Modified Euler's Method:-

$$y_1 = y_0 + h * f\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} * f(x_0, y_0)\right)$$

$$\vdots$$

$$y_n = y_{n-1} + h * f\left(x_{n-1} + \frac{h}{2}, y_{n-1} + \frac{h}{2} * f(x_{n-1}, y_{n-1})\right)$$

let $n=4$

$$\frac{dy}{dx} = 2y + 3e^x$$

$$y(0) = 0$$

$$y(0.2) = ?$$

$$h = \frac{x - x_0}{n} = \frac{0.2 - 0}{4} = 0.05$$

$$h = 0.05$$

$$h_2 = 0.025$$

$$y_1 = 0 + 0.05 * \left(3e^{0.025} + 2 \left[0 + 0.05 * (2 \times 0 + 3e^0) \right] \right) 0.05 \quad (x_1)$$

$$= 0.05 * (3.075945 + 0.15)$$

$$0.10 \quad (x_2)$$

$$0.15 \quad (x_3)$$

$$y_1 = 0.161297$$

$$x = h + x_0$$

$$0.20 \quad (x_4)$$

$$y_2 = 0.161297 + 0.05 * \left(3e^{0.075} + 2 \left[0.161297 + 0.025 * (3e^{0.05} + 2 * 0.161297) \right] \right)$$

$$y_2 = 0.161297 + 0.05 * \left[3e^{0.075} + 2 \left[0.161297 + 0.025 \right.$$

$$\left. * (2 \times 0.161297 + 3 \times e^{0.05}) \right] \right]$$

$$y_2 = 0.347800$$

$$y_3 = 0.347800 + 0.05 \left[3e^{0.125} + 2 \left[0.3478 + 0.025 \right. \right. \\ \left. \left. * (2 \times 0.3478 + 3 \times e^{0.1}) \right) \right]$$

$$y_3 = 0.56258$$

$$y_4 = 0.56258 + 0.05 \left[3e^{0.175} + 2 \left[0.56258 + 0.025 \right. \right. \\ \left. \left. * (2 \times 0.56258 + 3 \times e^{0.15}) \right) \right]$$

$$y_4 = 0.809052$$

Imp
 $\Rightarrow$ R-K Method (Runge-Kutta Method):
 $K = hf(x_0, y_0)$

Order 1:- Euler's Method $\Rightarrow y_1 = y_0 + K$

Order 2:- Modified Euler's Method $\Rightarrow y_1 = y_0 + K$
 $K = hf(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} * f(x_0, y_0))$

Order 3:- Runge's Method $\Rightarrow y_1 = y_0 + K$
 $K = \frac{1}{6} (K_1 + 4K_2 + K_3)$, $K_1 = hf(x_0, y_0)$
 $K_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_1)$, $K_3 = hf(x_0 + h, y_0 + K_1)$

Order 4

$$K' = hf(x_0 + h, y_0 + K_1)$$

V. Imp

Order 4: $y_1 = y_0 + K$
 $K = \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$, $K_1 = hf(x_0, y_0)$
 $K_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_1)$
 $K_3 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}K_2)$, $K_4 = hf(x_0 + h, y_0 + K_3)$

Que $\frac{dy}{dx} = 2y + 3e^x$ $y(0) = 0$ $y(0.2) = ?$ $n=4$

$$h = \frac{x - x_0}{n} = 0.05$$

$$\begin{bmatrix} x_0 = 0 \\ x_1 = 0.05 \\ x_2 = 0.10 \\ x_3 = 0.15 \\ x_4 = 0.20 \end{bmatrix}$$

$$y_1 = y_0 + K$$

$$K_1 = 0.05 * (2x_0 + 3e^0) = 0.15$$

$$K_2 = 0.05 \left(2 \times \left(0 + \frac{0.15}{2} \right) + 3e^{0.025} \right) = 0.161297$$

$$K_3 = 0.05 \left(2 \left(0 + \frac{0.161297}{2} \right) + 3e^{0.025} \right) = 0.161862$$

$$K_4 = 0.05 \left(2 \times \left(0 + 0.161862 \right) + 3e^{0.05} \right)$$

$$\boxed{K_4 = 0.173877}$$

$$K = \frac{1}{6} \left[0.15 + 2(0.161297) + 2(0.161862) + 0.173877 \right] = 0.161699$$

$$y_1 = y_0 + K$$

$$= 0 + 0.161699$$

$$\boxed{y_1 = 0.161699}$$

$$y_2 = y_1 + K$$

$$K_1 = hf(x_1, y_1)$$

$$= 0.05 * (2 * (.161699) + 3e^{0.05}) = 0.173860$$

$$K_2 = 0.05 * (2 * (.161699 + \frac{0.173860}{2}) + 3e^{(0.075)}) = .186546$$

$$K_3 = 0.05 * (2 * (.161699 + \frac{0.186546}{2}) + 3e^{0.075}) = .187179$$

$$K_4 = 0.05 * (2 * (.161699 + 0.187179) + 3e^{\frac{0.187}{0.075}}) = \cancel{.196570} \\ 0.200663$$

$$K = \frac{1}{6} [.173860 + 2(.186546) + 2(.187179) + .196570]$$

$$\boxed{K = 0.186313} \quad K = \cancel{+ .308} \cdot 186996$$

$$y_2 = 0.161699 + \cancel{0.186313} \cdot 186996$$

$$\boxed{y_2 = 0.348012} = 0.348695$$

$$y_3 = y_2 + K \quad K_1 = 0.05 * (2(0.348012) + 3e^{0.10}) = 0.200577$$

$$K_2 = 0.05 [2(0.348012 + \frac{0.200577}{2}) + 3e^{0.125}] = 0.214802$$

$$K_3 = 0.05 [2(0.348012 + \frac{0.214802}{2}) + 3e^{0.125}] = 0.215514$$

$$K_4 = 0.05 [2 [0.348012 + 0.215514] + 3e^{0.125}] = 0.226325$$

$$K = \frac{1}{6} [0.200577 + 2[.214802] + 2[.215514] + 0.226325]$$

$$\boxed{K = 0.214589}$$

$$K = \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4] = \cancel{0.246283} \quad 0.246283$$

$$y_4 = y_3 + K = 0.246283 + 0.562601$$

$$y_4 = 0.808884$$

$$y_3 = y_2 + K$$

$$K_1 = 0.05 * (2(0.348695) + 3e^{0.10}) = 0.200645$$

$$K_2 = 0.05 * (2(0.348695 + \frac{0.200645}{2}) + 3e^{0.125})$$

$$= 0.214874$$

$$K_3 = 0.05 * (2(0.348695 + \frac{0.214874}{2}) + 3e^{0.125})$$

$$= 0.215585$$

$$K_1 = 0.05$$

$$K_2 = 0.05$$

$$K_3 = 0.05$$

$$K_4 =$$

$$K = 1$$

$$K_4 = 0.05 * (3e^{.15} + 2(0.215585 +$$

$$= 0.230703$$

$$K = 0.215378$$

$$y_3 = y_2 + K$$

$$= 0.248681 + 0.215378 = 0.564073$$

$$\boxed{y_3 = 0.564073}$$

$$y_4 = y_3 + K$$

$$K_1 = 0.05 [2 * (0.564073) + 3e^{0.15}] = 0.230682$$

$$K_2 = 0.05 [2 * (0.564073 + \frac{0.230682}{2}) + 3e^{0.175}] = 0.246628$$

$$K_3 = 0.05 [2 * (0.564073 + \frac{0.246628}{2}) + 3e^{0.175}] = 0.247426$$

$$K_4 = 0.05 [2 * (0.564073 + \frac{0.247426}{2}) + 3e^{0.20}]$$

$$= 0.264360$$

$$K = \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4] = 0.247192$$

$$\boxed{y_4 = 0.811265}$$